Rigorous Formulation and Interval-Based Difference Array Computation of the Multi-Threshold Euler Characteristic

A Unified Framework for Topology, Valuations, and Linear-Time Binarization

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Abstract

Computing the topological Euler characteristic across multiple binarization thresholds is a core requirement in digital image analysis, shape filtering, and segmentation. While static evaluation on a binary complex is strictly local, repeatedly evaluating the Euler characteristic for G distinct threshold levels naively costs $\mathcal{O}(|A| \cdot G)$ operations. In this paper, we develop a comprehensive mathematical framework proving that: (1) the global Euler-Poincaré characteristic over cubical complexes decomposes into local valuations over dual Voronoi domains; (2) under grayscale thresholding t, the local quad functions $q_1(t), q_3(t), q_d(t)$ decompose into finite sums of interval indicator functions $\mathbf{1}_{[t_1,t_2]}(t)$; and (3) the global multi-threshold Euler characteristic curve $\chi(t)$ can be evaluated across all G levels in optimal $\mathcal{O}(|A| + G)$ time using sparse discrete difference arrays.

1 Topological Foundations on Cubical Complexes

Let $\mathbb{R}^N$ denote standard N-dimensional Euclidean space decomposed by the standard cubical complex $\mathcal{T}^N$.

Definition 1.1 (k-Cells of the Hypercubic Grid)**.** For each coordinate tuple $\mathbf{x} = (x_1, \ldots, x_N) \in \mathbb{Z}^N$ and binary vector $\mathbf{s} \in \{0,1\}^N$ with $\sum_{i=1}^N s_i = k$, the open k-cell $\sigma_k(\mathbf{x}, \mathbf{s})$ is:

$$\sigma_k(\mathbf{x}, \mathbf{s}) \coloneqq \prod_{i=1}^{N} I_i(x_i, s_i) \subset \mathbb{R}^N, \quad \text{where } I_i(x_i, s_i) \coloneqq \begin{cases} (x_i, x_i + 1) & \text{if } s_i = 1, \\ \{x_i\} & \text{if } s_i = 0. \end{cases} \tag{1}$$

Definition 1.2 (Foreground Complex and Euler Characteristic)**.** Let $I : \mathcal{C}_N(\mathcal{T}^N) \to \{0,1\}$ be a compactly supported binary image. The *foreground complex* $K(I) \subseteq \mathcal{T}^N$ is the topological closure of active N-cells:

$$K(I) \coloneqq \bigcup_{\substack{\sigma_N \in \mathcal{C}_N(\mathcal{T}^N) \\ I(\sigma_N)=1}} \bar{\sigma}_N. \tag{2}$$

The global Euler-Poincaré characteristic $\chi(K(I)) \in \mathbb{Z}$ is defined as:

$$\chi(K(I)) \coloneqq \sum_{k=0}^{N} (-1)^k n_k(K(I)), \tag{3}$$

where $n_k(K(I)) \coloneqq |\mathcal{C}_k(K(I))|$ is the number of active open k-cells in $K(I)$.

Theorem 1.3 (Additive Valuation Property)**.** *The Euler characteristic $\chi : \mathcal{K}(\mathcal{T}^N) \to \mathbb{Z}$ is an additive valuation on the distributive lattice of finite subcomplexes $\mathcal{K}(\mathcal{T}^N)$:*

$$\chi(A \cup B) = \chi(A) + \chi(B) - \chi(A \cap B), \quad \forall A, B \in \mathcal{K}(\mathcal{T}^N). \tag{4}$$

Proof. Let $C_k(K)$ denote the k-th cellular chain group of complex K. The indicator functions over sets of open k-cells satisfy:

$$\mathbf{1}_{\mathcal{C}_k(A\cup B)} = \mathbf{1}_{\mathcal{C}_k(A)} + \mathbf{1}_{\mathcal{C}_k(B)} - \mathbf{1}_{\mathcal{C}_k(A\cap B)}. \tag{5}$$

Taking the sum over all k-cells for each dimension $k \in \{0, \ldots, N\}$ and multiplying by $(-1)^k$ immediately yields the valuation identity. □

2 The Dual Voronoi Partition and Local Valuation

Definition 2.1 (Dual Voronoi Cell / Cell Group)**.** For each primal vertex $\mathbf{v} \in \mathcal{V}(\mathcal{T}^N) = \mathbb{Z}^N$, its *dual Voronoi cell* $C(\mathbf{v}) \subset \mathbb{R}^N$ is defined by:

$$C(\mathbf{v}) \coloneqq \left\{ \mathbf{x} \in \mathbb{R}^N \;\middle|\; \|\mathbf{x} - \mathbf{v}\|_\infty \leq \frac{1}{2} \right\} = \prod_{i=1}^{N} \left[v_i - \frac{1}{2}, v_i + \frac{1}{2} \right]. \tag{6}$$

Lemma 2.2 (Equipartition of k-Cells)**.** *Each open k-cell $\sigma_k \in \mathcal{C}_k(\mathcal{T}^N)$ is geometrically bounded by $V_k = 2^k$ vertices and is partitioned into V_k congruent subsectors across the dual cells $\{C(\mathbf{v}) \mid \mathbf{v} \in \mathcal{V}(\bar{\sigma}_k)\}$:*

$$\sum_{\mathbf{v}\in\mathbb{Z}^N} \mathbf{1}_{\{\sigma_k \cap \mathrm{int}(C(\mathbf{v})) \neq \emptyset\}} = V_k = 2^k. \tag{7}$$

Theorem 2.3 (Local Valuation Theorem)**.** *The global Euler characteristic is exactly localized to vertex contributions:*

$$\chi(K(I)) = \sum_{\mathbf{v}\in\mathbb{Z}^N} \Delta\chi(Q(\mathbf{v})), \quad \Delta\chi(Q(\mathbf{v})) \coloneqq \sum_{k=0}^{N} (-1)^k \frac{1}{2^k} n_k^{loc}(Q(\mathbf{v})), \tag{8}$$

where $Q(\mathbf{v}) \in \{0,1\}^{2^N}$ is the local state of the 2^N voxels meeting at vertex $\mathbf{v}$, and $n_k^{loc}(Q(\mathbf{v}))$ is the number of active k-cells incident to $\mathbf{v}$.

3 Grayscale Images and Indicator Interval Functions

3.1 Grayscale Level-Set Binarization

Let $\mathcal{A} : \mathbb{Z}^2 \to \{0, 1, \ldots, G-1\}$ denote a 2-dimensional discrete grayscale image with G discrete intensity levels.

For any threshold parameter $t \in [0, G-1]$, the thresholded binary image $I_t : \mathbb{Z}^2 \to \{0,1\}$ is defined pointwise:

$$I_t(x,y) \coloneqq \begin{cases} 1 & \text{if } \mathcal{A}(x,y) > t, \\ 0 & \text{if } \mathcal{A}(x,y) \leq t. \end{cases} \tag{9}$$

Under the 8-connected (W-model) convention, the Euler characteristic of the foreground complex at threshold t is given by Gray's formula:

$$\chi(t) = \frac{1}{4} \left(q_1(t) - q_3(t) - 2q_d(t) \right), \tag{10}$$

where $q_1(t), q_3(t), q_d(t)$ are the global counts of bit quads of type Q_1, Q_3, and Q_D present in I_t.

3.2 Local Quad Transitions as Interval Indicators

Let $\mathbf{v} \in \mathbb{Z}^2$ be an internal grid vertex, and consider the four grayscale intensities of the 2×2 quad surrounding $\mathbf{v}$:

$$\mathcal{Q}(\mathbf{v}) = \begin{bmatrix} g_{00} & g_{01} \\ g_{10} & g_{11} \end{bmatrix} \in \{0, \ldots, G-1\}^{2\times 2}. \tag{11}$$

Let (a, b, c, d) be the sorted permutation of these four scalar values such that:

$$a \leq b \leq c \leq d. \tag{12}$$

Theorem 3.1 (Indicator Interval Representation of Local Quads)**.** *For a given quad $\mathcal{Q}(\mathbf{v})$ with sorted intensity values $a \leq b \leq c \leq d$, its local contribution to the quad counters as a function of continuous threshold $t \geq 0$ is exactly expressed by indicator functions over integer intervals:*

$$\mathbf{1}_{Q_1(\mathbf{v})}(t) = \mathbf{1}_{[c,\, d-1]}(t) \quad \text{if } c < d, \tag{13}$$

$$\mathbf{1}_{Q_3(\mathbf{v})}(t) = \mathbf{1}_{[a,\, b-1]}(t) \quad \text{if } a < b, \tag{14}$$

$$\mathbf{1}_{Q_D(\mathbf{v})}(t) = \mathbf{1}_{[b,\, c-1]}(t) \quad \text{if } b < c \text{ and } \{c, d\} \text{ are diagonally positioned in } \mathcal{Q}(\mathbf{v}). \tag{15}$$

Proof. Recall that a pixel is active (1) if its intensity exceeds t. Thus:

1. **Type Q_1 (Exactly one 1):** Exactly one pixel is 1 if and only if the maximum intensity exceeds t while the remaining three do not: $d > t$ and $c \leq t$. For integer t, this is equivalent to $c \leq t \leq d-1$, which defines the interval $[c, d-1]$.

2. **Type Q_3 (Exactly three 1s):** Exactly three pixels are 1 if and only if the top three values exceed t while the minimum does not: $b > t$ and $a \leq t$. This is equivalent to $a \leq t \leq b-1$, defining the interval $[a, b-1]$.

3. **Type Q_D (Two diagonal 1s):** Exactly two pixels are 1 if and only if $c > t$ and $b \leq t$, yielding $t \in [b, c-1]$. This quad constitutes a diagonal configuration Q_D if and only if the two maximal elements $\{c, d\}$ occupy diagonally opposite positions in the 2×2 window. □

Corollary 3.2 (Global Quad Sums)**.** *The global counts $q_1(t), q_3(t), q_d(t)$ are sums of local indicator functions over all vertices $\mathbf{v}$:*

$$q_1(t) = \sum_{\mathbf{v}} \mathbf{1}_{Q_1(\mathbf{v})}(t), \quad q_3(t) = \sum_{\mathbf{v}} \mathbf{1}_{Q_3(\mathbf{v})}(t), \quad q_d(t) = \sum_{\mathbf{v}} \mathbf{1}_{Q_D(\mathbf{v})}(t). \tag{16}$$

4 Optimal Multi-Threshold Evaluation via Difference Arrays

4.1 The Difference Operator Representation

Evaluating $\chi(t)$ point-by-point across G threshold values naively requires updating arrays along interval spans, resulting in $\mathcal{O}(|A| \cdot G)$ complexity. By leveraging the algebraic structure of interval indicator functions, we transform interval updates into sparse point impulses.

Definition 4.1 (Discrete Difference Operator)**.** For any discrete sequence $f : \{0, \ldots, G-1\} \to \mathbb{Z}$, its forward difference sequence Δf is defined by:

$$\Delta f[t] \coloneqq f[t] - f[t-1] \quad (\text{with } f[-1] \coloneqq 0), \quad \text{so that } f[t] = \sum_{k=0}^{t} \Delta f[k]. \tag{17}$$

Proposition 4.2 (Sparse Boundary Impulses)**.** *The indicator function of an integer interval $[t_1, t_2]$ has a discrete difference sequence consisting of exactly two non-zero impulses:*

$$\Delta \mathbf{1}_{[t_1,t_2]}[t] = \delta[t - t_1] - \delta[t - (t_2 + 1)], \tag{18}$$

where $\delta[k]$ is the discrete Kronecker delta impulse ($\delta[0] = 1$ and $\delta[k \neq 0] = 0$).

Theorem 4.3 (Linear-Time Global Euler Difference Accumulator)**.** *Let $\Delta\chi_{diff} : \{0, \ldots, G\} \to \mathbb{Z}$ be a discrete difference array initialized to zero. For every vertex $\mathbf{v} \in \mathbb{Z}^2$ with sorted intensities $a \le b \le c \le d$:*

1. *If $c < d$: add $+1$ at index c, and add -1 at index d.*
2. *If $a < b$: add -1 at index a, and add $+1$ at index b.*
3. *If $b < c$ and $\{c, d\}$ are diagonal: add -2 at index b, and add $+2$ at index c.*

Then, for all thresholds $t \in \{0, \ldots, G-1\}$, the exact global Euler characteristic is obtained via a single prefix sum:

$$\chi(t) = \frac{1}{4} \sum_{k=0}^{t} \Delta\chi_{diff}[k]. \tag{19}$$

The entire curve $\{\chi(t)\}_{t=0}^{G-1}$ is computed in strictly $\mathcal{O}(|A| + G)$ time.

Proof. By Theorem 3.1 and linearity of the forward difference operator:

$$\begin{aligned} 4\Delta\chi[t] &= \Delta q_1[t] - \Delta q_3[t] - 2\Delta q_d[t] \\ &= \sum_{\mathbf{v}} \left(\Delta \mathbf{1}_{Q_1(\mathbf{v})}[t] - \Delta \mathbf{1}_{Q_3(\mathbf{v})}[t] - 2\Delta \mathbf{1}_{Q_D(\mathbf{v})}[t] \right). \end{aligned}$$

Substituting the two-point impulse representation of each interval:

$$\begin{aligned} \Delta \mathbf{1}_{Q_1(\mathbf{v})}[t] &= \delta[t-c] - \delta[t-d], \\ -\Delta \mathbf{1}_{Q_3(\mathbf{v})}[t] &= -\delta[t-a] + \delta[t-b], \\ -2\Delta \mathbf{1}_{Q_D(\mathbf{v})}[t] &= -2\delta[t-b] + 2\delta[t-c]. \end{aligned}$$

Accumulating these localized impulse weights into $\Delta\chi_{\text{diff}}$ over a single raster pass requires $\mathcal{O}(|A|)$ operations. The prefix sum integration runs in $\mathcal{O}(G)$ time. Thus, the total computational complexity is $\mathcal{O}(|A| + G)$. □

5 The Unified Fast Multi-Threshold Algorithm

Algorithm 1 presents the complete, single-pass algorithm computing the complete multi-threshold Euler characteristic spectrum.

Algorithm 1 Linear-Time Multi-Threshold Euler Characteristic Curve

Require: Grayscale image matrix $\mathcal{A}$ of size $H \times W$, number of gray levels G
Ensure: Array of Euler characteristics $\chi[0 \dots G-1]$

Allocate difference array $D[0 \dots G] \leftarrow 0$
Allocate result array $\chi[0 \dots G-1] \leftarrow 0$
$\mathcal{A}_{\text{pad}} \leftarrow \text{ZeroPad}(\mathcal{A}, \text{pad_width} = 1)$
for $y \leftarrow 0$ **to** H **do**
    for $x \leftarrow 0$ **to** W **do**
        Extract 4 neighbor pixels: $p_0 \leftarrow \mathcal{A}_{\text{pad}}[y, x],\ p_1 \leftarrow \mathcal{A}_{\text{pad}}[y, x+1]$
                                   $p_2 \leftarrow \mathcal{A}_{\text{pad}}[y+1, x],\ p_3 \leftarrow \mathcal{A}_{\text{pad}}[y+1, x+1]$
        Let (a, b, c, d) be the sorted values of $\{p_0, p_1, p_2, p_3\}$
        if $c < d$ **then** ▷ Contribution of Q_1
            $D[c] \leftarrow D[c] + 1$
            $D[d] \leftarrow D[d] - 1$
        end if
        if $a < b$ **then** ▷ Contribution of Q_3
            $D[a] \leftarrow D[a] - 1$
            $D[b] \leftarrow D[b] + 1$
        end if
        if $b < c$ **then** ▷ Contribution of Q_D
            if $(p_0, p_3 \in \{c, d\})$ **or** $(p_1, p_2 \in \{c, d\})$ **then**
                $D[b] \leftarrow D[b] - 2$
                $D[c] \leftarrow D[c] + 2$
            end if
        end if
    end for
end for
running_sum $\leftarrow 0$
for $t \leftarrow 0$ **to** $G-1$ **do** ▷ Integrate prefix sum
    running_sum $\leftarrow$ running_sum $+ D[t]$
    $\chi[t] \leftarrow$ running_sum$/4.0$
end for
return χ

6 Conclusion

We have established a rigorous, unified theory connecting abstract cellular valuations to practical real-time multi-threshold image analysis. By exploiting the equipartition of cubical cells over dual Voronoi domains and the piece-wise constant nature of threshold level-sets, local topological counts reduce to interval indicator functions. The forward difference operator maps these intervals to sparse discrete impulses, enabling the computation of the complete Euler characteristic spectrum $\chi(t)$ across all G gray levels in optimal $\mathcal{O}(|A| + G)$ time and $\mathcal{O}(G)$ auxiliary space.